Alex Tyh Wang

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Summary

Computer science undergraduate seeking a software engineering internship/co-op for Summer 2027, focused on data engineering and product analytics. Built Python evaluation pipelines, SQL-backed systems, and production AI products.

Education

Rochester Institute of Technology — Rochester, NY
Bachelor of Science in Computer Science — Expected May 2028

Skills

Languages: SQL, Python, C++, C#, Java, TypeScript, JavaScript
Data: Data modeling fundamentals, SQLite, PostgreSQL, Apache Spark, Hadoop/MapReduce
Evaluation: Data quality checks, data validation, deterministic Python testing, LLM evaluation
Engineering: Git, Linux CLI, REST APIs, asyncio, Jenkins, CI/CD, Angular, Next.js, .NET

Experience

Software Engineer Intern – AI Integrations — Jan 2026 – Aug 2026
Paychex — *Rochester, NY*

- Led development of an internal AI developer platform integrating **AI agents and MCP tooling** into enterprise workflows; presented the platform to executives.
- Built a custom **LLM evaluation harness** for developer and Sales workflows to compare model behavior, reliability, and usefulness.
- Reduced token usage by **55%** with a **subagent architecture** while improving internal agent workflow performance.

Student Researcher — Jan 2026 – May 2026
Rochester Institute of Technology — *Rochester, NY*

- Conducted research on **large language models and memory**.
- Focused on how memory and context shape long-term model behavior.

Software Engineer Intern — Feb 2025 – Aug 2025
Real Wave Capital — *Remote*

- Built production Angular and TypeScript interfaces for AI agents serving **100+ daily users**; rewrote tool-calling and context strategy to reduce token spend by roughly **50%**.
- Integrated OpenAI API and OpenRouter into .NET-backed agent workflows, improving endpoint response times by **25%** and supporting **6,000+ weekly requests**.

Projects

Chatio LLM Benchmark — Jun 2025 – Present
Python, Next.js, TypeScript — *chatio.dev*

- Built a **dataset of roughly 500 prompts** across five assistant-utility pillars; evaluated **30 leading models**.
- Built an **evaluation pipeline** combining blind human review, ensemble model scoring, and **deterministic Python validation** using regex and JSON parsing.
- Visualized model comparisons with radar charts; documented scoring and contamination controls in a public system card.

iMessages Chatbot Server — Nov 2025 – 2026
Python, SQL, SQLite, asyncio — *github.com/alextyhwang/iMessages-Chatbot-Server*

- Designed **SQLite data models** for message history and processing state, with a sender/timestamp index and parameterized **SQL** for conversation retrieval.
- Built incremental message ingestion using **SQL joins**, row-ID checkpoints, and null/empty-text filtering to process incoming messages and persist conversation history.
- Built a Python gateway for up to **20 concurrent conversations** using asyncio, debounce batching, and semaphore-limited calls.

Leadership and Activities

Founder and President — Jan 2022 – Present
Bay Area Stem Academy — *San Jose, CA | bayareastemacademy.org*

- Founded a 501(c)(3) nonprofit that taught **660+ students** through **15+ free camps**, partnered with **5+ schools**, and provided **$35,000+** in free STEM education.
- Led curriculum and a **SanDisk** volunteer partnership; donated **$5,000+** to partner organizations and teams.

Event Coordinator and Executive Board Member — Sep 2024 – Present
RIT Asian Culture Society — *Rochester, NY*

- Co-lead a **200+ member** community and help organize Night Market, an annual university-wide event with **600 attendees**.